DROTAVERINE

Non-anticholinergic

Slow IV

CHEMICAL NAME 1-[(3,4-diethoxyphenyl)methylidene]-6,7-diethoxy-1,2,3,4-tetrahydroisoquinoline (hydrochloride)

CLASS Musculotropic antispasmodic (phosphodiesterase-4 inhibitor, papaverine analogue)

TRADE NAMES No-Spa · Drotin · Spasmo-Drotin · Drotaverine HCl

THERAPEUTIC USES / INDICATIONS

- Smooth-muscle spasm: biliary and renal colic, GI spasm, IBS.
- Dysmenorrhoea.
- Spasm of the cervix during labour (some practices).

PHARMACODYNAMICS

- Inhibits **PDE-4** → ↑ cAMP in smooth muscle → relaxation.
- No anticholinergic effects → usable in glaucoma and BPH.

ADMINISTRATION

- **PO** tablets.
- **IM** or **IV** slowly (patient lying — hypotension risk).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, constipation.
- Headache, dizziness.
- Palpitations.

ADVERSE EFFECTS (SERIOUS)

- Hypotension (rapid IV).
- Anaphylaxis (injection contains sodium metabisulfite).
- AV block (rare).

DOSAGES

- **Adult PO:** 40–80 mg 2–3 times daily (max 240 mg/day).
- **SC/IM:** 40–240 mg/day in 1–3 doses; **renal/biliary colic:** 40–80 mg slowly IV (~30 s).
- **Children 6–12 yr:** max 80 mg/day in 2 doses; **>12 yr:** max 160 mg/day in 2–4 doses.

CONTRAINDICATIONS & CAUTIONS

- Severe hepatic, renal or cardiac failure; 2nd/3rd-degree AV block.
- Hypersensitivity (sulfite in injection).
- Porphyria; children <1 yr (injection).

KEY INTERACTIONS

- **Levodopa:** ↓ antiparkinsonian effect.
- **Other PDE inhibitors:** additive.

PHARMACOKINETICS

- **Onset:** IV 2–4 min; PO 45–60 min. **Duration:** 4–6 h.
- Metabolised in liver.
- Excreted in urine and bile. **t½:** 8–10 h.

NURSING MANAGEMENT

- IV: slowly with patient supine; monitor BP, HR.
- Assess colic pain response.
- Watch for allergic reactions in asthmatics.

PATIENT EDUCATION

- Report dizziness or fainting.
- Seek review for persistent abdominal pain.

KEY POINTS

- PDE-4 inhibitor antispasmodic.
- No anticholinergic effects.
- Renal/biliary colic.
- Slow IV — hypotension.

SOURCES CHECKED **[547]** Spaverin SwissPAR (Swissmedic) **[548]** No-Spa injection SmPC (Sanofi, Poland)

Full references p. 432–457